PRECALCULUS · VECTORS

Vector Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Vector Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. If $|u| = 3$ and $|v| = 4$, can $|u + v|$ be 7? Can it be 1? Can it be 8?

2. Find $\langle 2, -1 \rangle + \langle 3, 5 \rangle$.

2 Explore use the tool

1. Make $u = \langle 4, 3 \rangle$. Record its magnitude and direction.

2. Make $u + v = \langle 0, 0 \rangle$. How are u and v related?

3. Find vectors with $|u + v| = |u| + |v|$. What must be true about u and v?

4. Switch to $u - v$. Describe where $u - v$ appears on the picture.

3 Record use the tool and fill in the table

Operations

u	v	u + v	u − v	2u	|u + v|
⟨2, 1⟩	⟨1, 3⟩				
⟨−3, 2⟩	⟨3, −2⟩				
⟨4, 0⟩	⟨0, 3⟩				
⟨1.5, 2⟩	⟨3, 4⟩				

4 Explain what did you discover?

1. Add vectors by adding their _____.

2. $|u + v| \le$ _____, with equality only when u and v _____.

3. $k \cdot u$ multiplies the length by _____. If $k < 0$ it also _____.

4. $u - v = u +$ _____, and it goes from the tip of _____ to the tip of _____.

5. A plane flies with velocity $\langle 300, 0 \rangle$ km/h into a wind of $\langle -40, 30 \rangle$ km/h (wind speed 50 km/h). Explain why its ground speed is not $300 - 50$ or $300 + 50$.

5 Apply a new situation, no tool

1. Find $|\langle -5, 12 \rangle|$ and a unit vector in its direction.

2. $u = \langle 2, -3 \rangle$ and $v = \langle -1, 4 \rangle$. Find $3u - 2v$.

3. Find the direction angle of $\langle -3, 3 \rangle$.

4. A vector has magnitude 10 and direction 30°. Find its components.